The Nucleolus of a Characteristic Function Game
Author(s): David Schmeidler
Source: *SIAM Journal on Applied Mathematics*, Vol. 17, No. 6 (Nov., 1969), pp. 1163-1170
Published by: Society for Industrial and Applied Mathematics
Stable URL: https://www.jstor.org/stable/2099196
Accessed: 14-01-2026 20:46 UTC

JSTOR is a not-for-profit service that helps scholars, researchers, and students discover, use, and build upon a wide range of content in a trusted digital archive. We use information technology and tools to increase productivity and facilitate new forms of scholarship. For more information about JSTOR, please contact support@jstor.org.

Your use of the JSTOR archive indicates your acceptance of the Terms & Conditions of Use, available at https://about.jstor.org/terms

Society for Industrial and Applied Mathematics is collaborating with JSTOR to digitize, preserve and extend access to *SIAM Journal on Applied Mathematics*

This content downloaded from 154.59.125.71 on Wed, 14 Jan 2026 20:46:10 UTC
All use subject to https://about.jstor.org/terms

SIAM J. APPL. MATH.
Vol. 17, No. 6, November 1969

THE NUCLEOLUS OF A CHARACTERISTIC FUNCTION GAME*

DAVID SCHMEIDLER†

1. Introduction. A *characteristic function game* is a pair $[N, v]$ consisting of a set $N = \{1, 2, \cdots, n\}$ of n *players* and a *characteristic function* v, which maps every subset S of N, called a *coalition*, to a nonnegative number $v(S)$, called the *value* of coalition S. In addition, it is assumed that $v(S) = 0$ for each coalition S consisting of less than two players. An outcome of a game will be an n-tuple $x = (x^1, \cdots, x^n)$. A *payoff vector* is an outcome such that $x^i \geqq 0, i = 1, \cdots, n$, and $x(N) = v(N)$, where for each coalition S, $x(S)$ denotes $\sum_{i \in S} x^i$. X is the set of all payoff vectors.

In this paper we suggest an additional answer to the question of how the players of N divide the value of N among themselves, more precisely, which point of X is accepted as a compromise between the players. Given two payoff vectors, x, y, we describe a procedure for choosing the more acceptable among them. The number $v(S) - x(S)$, $S \subset N$, called the *excess* of S with respect to x, reflects the "attitude" of coalition S to the payoff vector x. The coalition which objects most strongly to a suggested payoff vector x is the one with the greatest excess. If $\max\{v(S) - x(S) | S \subset N\} > \max\{v(S) - y(S) | S \subset N\}$, we say that the payoff vector y is more acceptable than x. It raises less of an objection. If there is equality between the maxima, we compare the next greatest excess with respect to x and y, respectively. At this point it would be more convenient to give the formal definition.

For a fixed game $[N, v]$ and vector $x \in R^n$ let $\theta(x)$ be a vector in R^{2^n}, the components of which are the numbers $v(S) - x(S)$, arranged according to their magnitude, where S runs over the subsets of N; i.e., $i < j$ implies $\theta^i(x) \geqq \theta^j(x)$. The lexicographical[1] order on R^{2^n} induces a quasi-order on R^n.

Now we can describe completely the decision rule between any two outcomes. x is more acceptable than y if and only if $\theta(x) \prec \theta(y)$. Let Y be a subset of R^n; the *nucleolus* $\mathcal{N}(Y)$ of Y is the set of points in Y which are first in the quasi-order induced on Y. The nucleolus of the game, or simply nucleolus, is the nucleolus of the set of payoff vectors. In symbols:

$$\mathcal{N}(Y) = \{x \in Y | \theta(x) \precsim \theta(y) \text{ for all } y \in Y\}, \qquad \mathcal{N}(v) = \mathcal{N}(X).$$

The nucleolus of a game is closely related to other solution concepts of a characteristic function game. (The formal presentation of the following discussion is given in § 2.) We refer to the *bargaining set* $\mathcal{M}_1^{(i)}$, introduced by Aumann and Maschler [1], the *kernel* $\mathcal{K}$ of Davis and Maschler [2], and the *core*. There are certain relations between these sets. The core is included in the bargaining set. B. Peleg proved, using a fixed-point theorem, that for each game the bargaining set is

* Received by the editors December 3, 1968.

† Department of Mathematics, The Hebrew University of Jerusalem, Jerusalem, Israel. This research was sponsored in part by the Logistics and Mathematical Statistics Branch, Office of Naval Research, Washington, D.C., under Contract F61052 67 C 0094 and Contract N62558-4355.

[1] $\theta(x) \prec \theta(y)$ implies $\vee_{i=1}^{2^n}[(\theta^i(x) < \theta^i(y))$ and $\wedge_{j=1}^{i-1}(\theta^j(x) = \theta^j(y))]$ where $\wedge$ denotes conjunction and $\vee$ disjunction.

This content downloaded from 154.59.125.71 on Wed, 14 Jan 2026 20:46:10 UTC
All use subject to https://about.jstor.org/terms

nonempty [5]. The kernel of a game is a subset of the bargaining set. Its existence was proved, at first, in the same way as Peleg's proof for $\mathscr{M}_1^{(i)}$. Maschler and Peleg [4] gave an alternative, algebraic proof of the existence of the kernel. In the same paper they showed that the kernel meets the core whenever the core is nonempty.

We prove that the nucleolus is included in this intersection, and that it is a subset of the kernel. Thus, in a sense, the nucleolus is the third stage in the development of the bargaining theory of Aumann and Maschler, coming after $\mathscr{M}_1^{(i)}$ and $\mathscr{K}$.

The idea of the proof of the existence of the nucleolus is closely related to its definition and implicitly appears in the proof of Theorem 5.2 in [4]. Via nucleolus one has the most simple and elementary existence proof of the kernel and the bargaining set, actually two existence proofs, one topological and the other algebraic. Moreover, one can generalize the definition of the kernel and prove its existence for an arbitrary compact convex (nonempty) set of outcomes.

One of the most appealing properties of the nucleolus, as a solution concept, is its uniqueness. Another interesting result proved in the paper is the continuity of $\mathscr{N}(v)$ as a function from games to payoffs.

Results concerning the nucleoli of special classes of games were obtained by others. B. Peleg [6] investigated the connection between the weights of a constant sum majority game and its nucleolus. A. Kopelowitz [3] computed the nucleoli of all 6 and 7 person weighted majority constant sum games and all 6 person superadditive weighted majority nonconstant sum games. He also prepared a computer program for computing the nucleoli of an arbitrary 10 (or less) person game.

2. Main results.

THEOREM 1. *Every nonempty compact subset of R^n has a nonempty nucleolus.*

The proof of this theorem makes use of elementary topology. When the subset in consideration is a compact polytope, we also give an algebraic proof.

THEOREM 2. *The nucleolus of a convex set consists of one outcome at most.*

Before stating the next result, let us begin with several definitions. A *coalition structure* is a partition[2] of N. A payoff configuration is a coalition structure $\tau = (S_1, \cdots, S_k)$ together with an outcome x such that $x^i \geqq 0, i = 1, \cdots, n$, $x(S_j) = v(S_j), j = 1, \cdots, k$. Y_τ will denote the set of all such outcomes for a fixed τ. For each pair of distinct players, i, j and an outcome x, set

$$s_{ij}(x) = \max\{v(S) - x(S) | S \subset N,\ i \in S, j \notin S\}.$$

We say that player i *outweighs* player j if they both belong to the same coalition S_l of τ, $x^j > 0$ and $s_{ij}(x) > s_{ji}(x)$. The *kernel* of the coalition structure τ is the set of vectors in Y_τ such that for each pair of players (i, j), i *does not* outweigh j. The kernel is a subset of the bargaining set $\mathscr{M}_1^{(i)}$.

For the use of this paper we shall define the kernel of an arbitrary subset Y of R^n. We say that player i *outweighs* player j, via outcome x, if $s_{ij}(x) > s_{ji}(x)$ and there is an $\varepsilon_0 > 0$ such that for every ε, $0 < \varepsilon \leqq \varepsilon_0$, the outcome x_ε belongs to Y, where

$$x_\varepsilon^t = \begin{cases} x^t + \varepsilon, & t = i, \\ x^t - \varepsilon, & t = j, \\ x^t, & \text{otherwise}. \end{cases}$$

[2] A partition of N is a collection of nonempty, pairwise disjoint coalitions, the union of which is N.

This content downloaded from 154.59.125.71 on Wed, 14 Jan 2026 20:46:10 UTC
All use subject to https://about.jstor.org/terms

The *kernel* $\mathscr{K}(Y)$ of Y is the set of vectors in Y such that, for each pair of distinct players (i, j), i does not outweigh j. Of course, the new definition of the kernel coincides with the original one for the sets Y_τ.

THEOREM 3. *For every subset Y of R^n, $\mathscr{N}(Y) \subset \mathscr{K}(Y)$.*

Neither of the existence proofs of the kernel mentioned in the Introduction can be generalized in order to be applied to the kernel as defined above. However, Theorems 1 and 3 insure that $\mathscr{K}(Y)$ is nonempty for an arbitrary compact (nonempty) set Y.

One can also generalize the definition of the bargaining set $\mathscr{M}_1^{(i)}$ in the same manner (and prove that it contains the kernel for each Y.)

The *core* of the game is the set of payoff vectors in X such that $v(S) - x(S) \leqq 0$ for every nontrivial coalition ($v(N) = x(N)$ and $v(\varnothing) = x(\varnothing)$ for all x in X). More generally, the ε-core, ε a real number, is the set of payoff vectors in X such that $v(S) - x(S) \leqq \varepsilon$ for every nontrivial coalition. The core is the 0-core.

THEOREM 4. *$\mathscr{N}(v)$ belongs to every nonempty ε-core.*

The proof of this theorem derives directly from the definition of the nucleolus and is therefore not mentioned in the sequel.

Let us look at the set V of all the n-person characteristic function games. Fix once and for all an order on the coalitions. Every game can be uniquely presented as a vector in R^{2^n} by

$$v^i = v(S_i).$$

Henceforward we shall not distinguish between a game and its presentation. The set V as a subset of R^{2^n} is a topological space and $\mathscr{N}(v)$ maps vectors in V to vectors in R^n.

THEOREM 5. *The nucleolus is a continuous function of the characteristic function.*

In the same manner, the kernel of the game is a set-valued function from V to R^n, and we get the result that the kernel has a continuous skeleton.

The proof of Theorem 5 is based on an alternative definition of the nucleolus Let us begin with a new binary relation $*\prec$ defined on the set of the outcomes. (Do not confuse this with the relation $\prec$ introduced earlier.)

We say that x is star preferred to y, $x *\prec y$, if and only if

$$\max_{\{S \mid y(S) > x(S)\}} v(S) - x(S) < \max_{\{S \mid x(S) > y(S)\}} v(S) - y(S).$$

This means that the objection to the outcome x by the coalitions which prefer y to x is smaller than the objection to the outcome y by the coalitions which prefer x to y.

Similarly we define the relations $x \overset{*}{\sim} y$ and $x *\precsim y$. The definitions imply that for each pair of outcomes (x, y) we have $x *\prec y$ or $x \overset{*}{\sim} y$ or $y *\prec x$.

The equivalence of the definitions, in case of compact convex sets, is implied by the following lemma.

LEMMA. *Given a compact convex set Y, $\mathscr{N}(Y)$ is star preferred to any other outcome in Y.*

3. Proof of Theorem 1.

Claim 1. *$\theta^i(x)$ is a continuous function of x, $i = 1, \cdots, 2^n$.*

$v(S) - x(S)$ is a continuous function of x. We give an equivalent definition

This content downloaded from 154.59.125.71 on Wed, 14 Jan 2026 20:46:10 UTC
All use subject to https://about.jstor.org/terms

of $\theta^i(x)$:

$$\theta^i(x) = \max\{\min\{v(S) - x(S) | S \in \underset{\sim}{A}\} | \underset{\sim}{A} \subset 2^N, \overline{\overline{\underset{\sim}{A}}} = i\}.$$

Using the known result that a function defined as maximum or minimum of a finite number of continuous functions is continuous, we get the continuity of $\theta^i(x)$, $i = 1, \cdots, 2^n$.

Next let us define

$$Y_1 = \{x \in Y | \theta^1(x) \leqq \theta^1(y) \text{ for all } y \in Y\}$$

$$Y_i = \{x \in Y_{i-1} | \theta^i(x) \leqq \theta^i(y) \text{ for all } y \in Y_{i-1}\}, \qquad i = 2, \cdots, 2^n.$$

Claim 2. *Y_i is compact and nonempty, $i = 1, \cdots, 2^n$.*

Using the compactness of Y and the continuity of $\theta^i(x)$ for each i we easily prove the assertion.

Obviously $Y_{2^n} = \mathcal{N}(Y)$, so Theorem 1 is proved.

Algebraic proof. The proof is based on the next (algebraic) claim.

If $f(x)$ is a linear function in a compact, convex, polytope, then it attains its maximum (minimum) in a compact, convex subpolytope (which may consist of a unique vertex).

Let $\{S_i\}$, $i = 1, \cdots, 2^n$, be an ordered set of coalitions. Then we define:

$$W = \{x \in R^n | i < j \text{ implies } v(S_i) - x(S_i) \geqq v(S_j) - x(S_j)\}.$$

Claim 3. *If Y is a nonempty intersection of a compact, convex polytype with W, then $\mathcal{N}(Y) \neq \varnothing$.*

W is closed and convex, therefore Y is a compact, convex polytype included in W. For each $x \in W$, $\theta^i(x) = v(S_i) - x(S_i)$ and this is a linear function of x in W. We define

$$Y_1 = \{x \in Y | \text{ for all } y \in Y, \theta^1(x) \leqq \theta^1(y)\},$$

$$Y_i = \{x \in Y_{i-1} | \text{ for all } y \in Y_{i-1}, \theta^i(x) \leqq \theta^i(y)\}, \qquad i = 2, \cdots, 2^n.$$

Each Y_i is a compact, convex subpolytope of Y_{i-1} and $Y_{2^n} = \mathcal{N}(Y)$.

There is a finite number of possible orders on coalitions ($2^n!$), and for each order we define the set W. Given a compact polytope Y in R^n, it can be expressed as a *finite* union of compact, convex, nonempty polytopes Y_j, each of them included in only one set W. Let $y_j \in \mathcal{N}(Y_j)$, and $\theta(y_{j_0}) \precsim \theta(y_j)$ for all j; then $y_{j_0} \in \mathcal{N}(Y)$.

4. Proof of Theorem 2. It is enough to prove that $\theta(x) = \theta(y)$ and $x \neq y$ imply

$$\theta(\tfrac{1}{2}(x + y)) \prec \theta(x).$$

Let $x, y \in R^r$ and $\eta : R^r \to R^r$ be the ordering function:

$$\eta^i(z) = \max\{\min\{z^j | j \in A\} | A \subset \{1, 2, \cdots, r\} \text{ and } \overline{\overline{A}} = i\}.$$

An equivalent definition of η is: $\eta^1(z) = \max\{z^t\}$; if we have defined $\eta^1(z) = z^{j_1}, \cdots, \eta^i(z) = z^{j_i}$, then $\eta^{i+1}(z) = \max\{z^t | t \neq j_1, \cdots, j_i\}$.

Claim 4.

$$\eta(x + y) \precsim \eta(x) + \eta(y).$$

This content downloaded from 154.59.125.71 on Wed, 14 Jan 2026 20:46:10 UTC
All use subject to https://about.jstor.org/terms

Let $\eta^t(x) = x^{i_t}$, $\eta^t(y) = y^{j_t}$, $\eta^t(x+y) = x^{k_t} + y^{k_t}$, $t = 1, \cdots, r$. Obviously $x^{k_1} \leqq x^{i_1}$ and $y^{k_1} \leqq y^{j_1}$, so

$$x^{k_1} + y^{k_1} \leqq x^{i_1} + y^{j_1}.$$

If equality holds, then $x^{k_1} = x^{i_1}$ and $y^{k_1} = y^{j_1}$. Otherwise, when inequality holds, $\eta(x+y) \prec \eta(x) + \eta(y)$. In the former case, on reasoning by induction on r, the proof is completed. Moreover, if $\eta(x+y) = \eta(x) + \eta(y)$, then $x^{i_t} = x^{k_t}$ and $y^{j_t} = y^{k_t}$, $t = 1, \cdots, r$.

We shall now return to R^{2^n}. Our vectors are $\{v(S) - x(S)\}_{S \subset N}$, $\{v(S) - y(S)\}_{S \subset N}$ and $\{2v(S) - x(S) - y(S)\}_{S \subset N}$. It can easily be seen that

$$\eta(\{v(S) - x(S)\}_{S \subset N}) = \theta(x),$$

$$\eta(\{v(S) - y(S)\}_{S \subset N}) = \theta(y),$$

$$\eta(\{2v(S) - x(S) - y(S)\}_{S \subset n}) = 2\theta(\tfrac{1}{2}(x+y)).$$

According to our claim, we have $2\theta(\frac{1}{2}(x+y)) \precsim \theta(x) + \theta(y)$. If $2\theta(\frac{1}{2}(x+y)) \prec \theta(x) + \theta(y)$, the proposition is proved. Otherwise, $2\theta(\frac{1}{2}(x+y)) = \theta(x) + \theta(y)$; and if $2\theta^t(\frac{1}{2}(x+y)) = v(S_{k_t}) - x(S_{k_t}) + v(S_{k_t}) - y(S_{k_t})$, $\theta^t(x) = v(S_{i_t}) - x(S_{i_t})$ and $\theta^t(y) = v(S_{j_t}) - y(S_{j_t})$, then for $t = 1, \cdots, 2^n$, $v(S_{k_t}) - x(S_{k_t}) = v(S_{i_t}) - x(S_{i_t})$ and $v(S_{k_t}) - y(S_{k_t}) = v(S_{j_t}) - y(S_{j_t})$. Using the assumption $\theta(x) = \theta(y)$, we have $v(S_{k_t}) - x(S_{k_t}) = v(S_{k_t}) - y(S_{k_t})$ for each t, or equivalently, $x(S) = y(S)$ for each coalition S. The last equation implies that $x = y$, contrary to the assumption.

Remark. Theorem 2 holds even if some of the coalitions are forbidden, i.e., the characteristic function v is defined only on a subset of 2^N. The condition is that this subset should include a collection of n coalitions, the characteristic vectors of which are linearly independent.

5. Proof of Theorem 3. Let $x \in \mathcal{N}(Y)$. If x does not belong to the kernel, there are two players i, j and an $\varepsilon_0 > 0$ such that $s_{ij}(x) > s_{ji}(x)$ and $x_\varepsilon \in Y$, $0 < \varepsilon \leqq \varepsilon_0$. Define

$$\delta = \tfrac{1}{2} \min \{\varepsilon_0, s_{ij}(x) - s_{ji}(x)\}.$$

Claim 5.

$$\theta(x_\delta) \prec \theta(x).$$

Let k be the cardinal number of $\underset{\sim}{K}$ where

$$\underset{\sim}{K} = \{S | v(S) - x(S) \geqq s_{ij}(x) \text{ and } i \in S \text{ implies } j \in S\}.$$

$\theta^k(x) \geqq s_{ij}(x)$ but $\theta^{k+1}(x) = s_{ij}(x)$ because there is at least one coalition R such that $i \in R$, $j \notin R$ and $v(R) - x(R) = s_{ij}(x)$ (which is what $s_{ij}(x)$ means). Of course, for such a coalition R, $v(R) - x_j(R) = s_{ij}(x) - \delta$. By the transfer of δ from j to i the excess of coalitions containing j but not i is increased to at most $s_{ji}(x) + \delta$, which is still less than $s_{ij}(x)$. So $\theta^{k+1}(x_\delta) < s_{ij}(x)$. On the other hand, the excess of coalitions in K is not changed by this transfer.

The claim we have just proved contradicts the initial assumption.

6. Proof of Theorem 5. We begin with the proof of the lemma, for which the following two claims are needed.

This content downloaded from 154.59.125.71 on Wed, 14 Jan 2026 20:46:10 UTC
All use subject to https://about.jstor.org/terms

Claim 6. *$x \,{}^*\!\precsim y$ implies $\alpha x + \beta y \,{}^*\!\prec y$ when $x \neq y$ and $0 < \alpha = 1 - \beta < 1$.*
Let

$$\underset{\sim}{A} = \{S | x(S) > y(S)\}, \qquad \underset{\sim}{B} = \{S | y(S) > x(S)\};$$

so

$$\{S | \alpha x + \beta y(S) > y(S)\} = \underset{\sim}{A},$$

$$\{S | y(S) > \alpha x + \beta y(S)\} = \underset{\sim}{B}.$$

$x \,{}^*\!\precsim y$ means, by definition,

$$\max_{S \in \underset{\sim}{B}} v(S) - x(S) \leqq \max_{S \in \underset{\sim}{A}} v(S) - y(S). \tag{1}$$

Let

$$p = \max_{S \in \underset{\sim}{B}} v(S) - \alpha x + \beta y(S),$$

$$q = \max_{S \in \underset{\sim}{A}} v(S) - y(S).$$

We have to prove that $p < q$; but q is equal to the right side of the inequality (1), and p is smaller than its left side, because for each S in $\underset{\sim}{B}$,

$$\alpha x + \beta y(S) > x(S).$$

Claim 7. *$x \prec y$ implies $x \,{}^*\!\precsim y$.*

$x \prec y$ means, by definition, that there is an i_0 such that $\theta^i(x) = \theta^i(y)$ for all $i < i_0$ and $\theta^{i_0}(x) < \theta^{i_0}(y)$. Let us cancel all the components of $\theta(x)$ and $\theta(y)$ such that $x(S) = y(S)$.

Case (a). All the components of the vector $\theta(x)$ preceding $\theta^{i_0}(x)$ were canceled. For each such component $\theta^i(x)$ there is a coalition R such that $\theta^i(x) = v(R) - x(R)$. But $x(R) = y(R)$, so there is a component of $\theta(y)$ equal to $\theta^i(x)$ which is also canceled. As a conclusion we have that the first $i_0 - 1$ components of $\theta(y)$ were also canceled and $\theta^{i_0}(y)$ has not been canceled. Hence there is a coalition T with $x(T) \neq y(T)$ and $\theta^{i_0}(y) = v(T) - y(T)$. Obviously $x(T) > y(T)$, because otherwise

$$v(T) - x(T) > v(T) - y(T) = \theta^{i_0}(y) > \theta^{i_0}(x).$$

But all the components of $\theta(x)$ which are greater than $\theta^{i_0}(x)$ have been canceled, a contradiction. Hence we have in Case (a):

$$\theta^{i_0}(y) = v(T) - y(T) = \max_{\{R | x(R) > y(R)\}} v(R) - y(R) > \theta^{i_0}(x)$$

$$\geqq \max_{\{R | x(R) \neq y(R)\}} v(R) - x(R) \geqq \max_{\{R | y(R) > x(R)\}} v(R) - x(R),$$

which means $x \,{}^*\!\prec y$.

Case (b). Not all the components preceding $\theta^{i_0}(x)$ were canceled. Let S be such that $v(S) - x(S)$ is the greatest uncanceled component of $\theta(x)$. The greatest uncanceled component $v(R) - y(R)$ of $\theta(y)$ is equal to $v(S) - x(S)$. (Because $i < i_0$ implies $\theta^i(x) = \theta^i(y)$.) As in Case (a) we prove that $y(S) > x(S)$ and $x(R) > y(R)$, and this implies $x \,{}^*\!\precsim y$.

COROLLARY. *$x \precsim y$ implies $x \,{}^*\!\precsim y$.*

It is enough to show that $x \sim y$ implies $x \,{}^*\!\precsim y$. But this implication is trivial when we observe that $x \sim y$ implies, by definition, Case (b).

This content downloaded from 154.59.125.71 on Wed, 14 Jan 2026 20:46:10 UTC
All use subject to https://about.jstor.org/terms

Case (b) also indicates how to construct a counterexample to the implication: $x \prec y$ implies $x *\prec y$.

Now we complete the proof of the lemma. Let x denote the nucleolus of the set Y and assume that there is an outcome y, $y \neq x$ and $y *\precsim x$. Let $z = \frac{1}{2}(x + y)$; then by Claim 6, $z *\prec x$. But using the uniqueness of the nucleolus (Theorem 2) we have $x \prec z$, and by Claim 7, $x *\precsim z$, a contradiction.

Proof of the continuity (Theorem 5). Let $v_n \to v_0$, $x_n \to x_0$, and for $n \geqq 1$, let x_n be the nucleolus of v_n. The assumption that x_0 is not the nucleolus of v_0 implies the existence of a payoff y such that $y *\prec x_0$ in v_0, i.e.,

$$\max_{\{S \mid x_0(S) > y(S)\}} v_0(S) - y(S) < \max_{\{S \mid y(S) > x_0(S)\}} v_0(S) - x_0(S). \tag{2}$$

We shall define a sequence of payoffs y_n such that for a great enough integer n, $y_n *\prec x_n$ in v_n, and this will contradict the fact that x_n is the nucleolus of v_n. We define $y_n = y + x_n - x_0$ for $n \geqq 1$. Of course $y_n \to y$, and for each coalition S we have:

$$x_n(S) = y_n(S) \quad \text{if and only if} \quad x_0(S) = y(S),$$

$$x_n(S) > y_n(S) \quad \text{if and only if} \quad x_0(S) > y(S),$$

$$x_n(S) < y_n(S) \quad \text{if and only if} \quad x_0(S) < y(S).$$

Because of the strong inequality in (2) we have, for a great enough integer n:

$$\max_{\{S \mid x_n(S) > y_n(S)\}} v_n(S) - y_n(S) < \max_{\{S \mid y_n(S) > x_n(S)\}} v_n(S) - x_n(S)$$

and that means $y_n *\prec x_n$ in v_n.

To complete the proof we have to show that y_n is a payoff. (A payoff is an individually rational and pareto optimal outcome, i.e., $x^i \geqq 0$ and $x(N) = v(N)$.) If for a player i, $y^i = 0$ and $x_0^i > x_n^i$, we have $y_n^i < 0$, which means that y_n is not a payoff. But $x_0^i > x_n^i$ implies $x_0^i > 0$. Let $z = \frac{1}{2}(y + x_0)$; then applying Claim 6, $z *\prec x_0$, and for each i with $x_0^i > 0$, $z^i > 0$. Hence we may assume that $y^i > 0$ or $x_0^i \leqq x_n^i$, which ensures that y_n, for great enough n, is a payoff. The problem which appears when $y^i = v(N)$ and $x_0^i < x_n^i$ can be solved in the same manner.

Acknowledgments. The author is indebted to Professors R. J. Aumann and M. Maschler for several helpful suggestions. Especially, the second proof of Theorem 1 and a simplification in the proof of Theorem 2 were suggested by Professor Aumann.

REFERENCES

[1] R. J. Aumann and M. Maschler, *The bargaining set for cooperative games*, Advances in Game Theory, M. Dresher, L. S. Shapley and A. W. Tucker, eds., Annals of Mathematical Studies, no. 52, Princeton University Press, Princeton, 1964, pp. 443–476.

[2] M. Davis and M. Maschler, *The kernel of a cooperative game*, Naval Res. Logist. Quart., 12 (1965), pp. 223–259.

[3] A. Kopelowitz, *Computation of the kernels of simple games and the nucleolus of n-person games*, RM 37, Research Program in Game Theory and Mathematical Economics, The Hebrew University of Jerusalem, Jerusalem, 1967.

[4] M. Maschler and B. Paleg, *A characterization, existence proof and dimension bounds for the kernel of a game*, Pacific J. Math., 18 (1966), pp. 289–328.

This content downloaded from 154.59.125.71 on Wed, 14 Jan 2026 20:46:10 UTC
All use subject to https://about.jstor.org/terms

[5] B. PALEG, *Existence theorem for the bargaining set* $\mathscr{M}_1^{(i)}$, Essays in Mathematical Economics in Honor of Oskar Morgenstern, M. Shubik, ed., Princeton University Press, Princeton, 1967, pp. 53–56. Also: Bull. Amer. Math. Soc., 69 (1963), pp. 109–110.

[6] ———, *On weights of constant-sum majority games*, this Journal, 16 (1968), pp. 527–532.

This content downloaded from 154.59.125.71 on Wed, 14 Jan 2026 20:46:10 UTC
All use subject to https://about.jstor.org/terms